

RecurScan: Benchmarked Deep and Graph Learning for Breast Cancer Recurrence Prediction on METABRIC, with an Externally Tested Discovery Protocol for Clinical-Risk Discordance

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Abstract

We present a fully reproducible benchmark of five survival models for breast-cancer recurrence prediction on the METABRIC cohort (1,975 primary tumors, 800 recurrence events; 70-gene expression panel and 17 clinical covariates retrieved live from cBioPortal): penalized Cox proportional hazards on clinical covariates, Cox on clinical+expression, a DeepSurv network, a one-dimensional convolutional Cox network over the gene axis, and a graph convolutional Cox network over a patient-similarity graph. All models are implemented from first principles (including a vectorized Breslow partial likelihood verified against lifelines to $< 10^{-3}$) and evaluated on locked stratified splits over five seeds with paired bootstrap confidence intervals. Penalized Cox on clinical+expression achieves the strongest held-out concordance, 0.6801 ± 0.0052 ; deep models trail (0.641–0.660), an honest negative we analyze. The fitted baseline stratifies held-out patients into risk terciles separated at log-rank $p = 7.9 \times 10^{-16}$. We then apply a pre-registered discovery protocol to clinical-risk discordance: among low-NPI patients, early recurrences (≤ 60 months) show significant up-regulation of eleven mitotic genes (Benjamini–Hochberg q down to 8.8×10^{-5}), but the signature fails both multivariable prediction (nested-CV AUC = 0.50) and external replication on TCGA-BRCA ($p = 0.36$), and is reported as a bounded negative. All code, tests, data snapshots, and figures are released with the paper.

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1 Introduction

Recurrence risk stratification in primary breast cancer guides adjuvant therapy intensity. Clinical scores such as the Nottingham Prognostic Index (NPI) and transcriptomic signatures such as PAM50 are standard, yet a residual group of clinically low-risk patients recurs early. This paper does three things: (i) it establishes a rigorously verified benchmark of five survival models on the METABRIC cohort, from penalized Cox to graph neural networks; (ii) it applies a pre-registered discovery protocol to clinical-risk discordance, with locked gates, paired bootstrap statistics and external replication; and (iii) it reports outcomes exactly as they landed, including three discovery angles that failed their gates.

2 Data

2.1 Cohort

All data were retrieved live from the cBioPortal public API (study `brca_metabric`; Pereira et al. 2016, Curtis et al. 2012). Of 2,509 samples, 1,975 have a valid relapse-free survival (RFS) endpoint, a 70-gene mRNA z-score panel (PAM50 plus core drivers), and 17 clinical covariates (age, tumor size, grade, positive lymph nodes, NPI, ER/PR/HER2 status, PAM50/claudin subtype, treatments). The endpoint is RFS: 800 events (40.5%), median follow-up 140.9 months among censored patients.

2.2 External replication cohort

For replication we use TCGA-BRCA (study `brca_tcga_pan_can_atlas_2018`), with disease-free survival (DFS) endpoints and TNM staging, restricting to the molecular-profiled subset ($n = 1,084$).

3 Methods

3.1 The Cox model and its partial likelihood

Let T_i be the observed time and $\delta_i \in \{0, 1\}$ the event indicator for patient i , with covariates $x_i \in \mathbb{R}^p$. The Cox proportional hazards model posits

$$\lambda(t | x_i) = \lambda_0(t) \exp(\eta_i), \quad \eta_i = x_i^\top \beta. \quad (1)$$

Definition 1 (Risk set). $\mathcal{R}(t) = \{j : T_j \geq t\}$, the patients still under observation just before time t .

Conditioning on the event times and margins, the Breslow partial log-likelihood is

$$\ell(\beta) = \sum_{i: \delta_i=1} \left[\eta_i - \log \sum_{j \in \mathcal{R}(T_i)} \exp(\eta_j) \right]. \quad (2)$$

Proposition 1 (Score of the partial likelihood).

$$\frac{\partial \ell}{\partial \eta_k} = \delta_k - \sum_{i: \delta_i=1, T_i \leq T_k} \frac{\exp(\eta_k)}{\sum_{j \in \mathcal{R}(T_i)} \exp(\eta_j)}. \quad (3)$$

Proof sketch. Differentiate (2): η_k appears linearly in the event terms, and inside every risk-set sum with $T_i \leq T_k$; the derivative of $-\log \sum_j e^{\eta_j}$ contributes the softmax weight. $\square$

Vectorized evaluation. Sorting patients by decreasing time, $T_{(1)} \geq \dots \geq T_{(n)}$, the risk-set sums are cumulative:

$$\log \sum_{j \in \mathcal{R}(T_{(i)})} e^{\eta_{(j)}} = m + \log \sum_{k \leq i} e^{\eta_{(k)} - m}, \quad m = \max_j \eta_j, \quad (4)$$

so (2) is $O(n \log n)$. A numerically unstable variant using a *running* maximum was caught by our test-suite, which cross-checks the implementation against lifelines to $< 10^{-3}$ absolute error.

3.2 Baseline hazard and the logrank test

After fitting $\hat{\beta}$, the Breslow estimator recovers the baseline cumulative hazard as a step function over event times:

$$\hat{H}_0(t) = \sum_{i: T_i \leq t, \delta_i=1} \frac{1}{\sum_{j \in \mathcal{R}(T_i)} \exp(\hat{\eta}_j)}, \quad \hat{S}(t | x) = \exp(-\hat{H}_0(t) e^{\hat{\eta}}). \quad (5)$$

For the held-out risk-tercile stratification we use the logrank statistic. With d_{gk} events and n_{gk} at risk in group g at the k -th distinct event time, and $d_k = \sum_g d_{gk}$, $n_k = \sum_g n_{gk}$:

$$Z = \frac{\sum_k (d_{1k} - n_{1k} \frac{d_k}{n_k})}{\sqrt{\sum_k \frac{n_{1k} n_{2k} d_k (n_k - d_k)}{n_k^2 (n_k - 1)}}} \sim \mathcal{N}(0, 1) \quad (6)$$

under the null of equal hazards; (6) yields the $p = 7.9 \times 10^{-16}$ stratification result in Section 4. Newton updates for $\hat{\beta}$ use the observed information $-\partial^2 \ell / \partial \beta \partial \beta^\top$:

$$\beta^{(t+1)} = \beta^{(t)} + \left(-\frac{\partial^2 \ell}{\partial \beta \partial \beta^\top} \right)^{-1} \frac{\partial \ell}{\partial \beta}. \quad (7)$$

3.3 Concordance index

Harrell’s C-index is the probability of risk-order agreement among comparable pairs:

$$C = \frac{\sum_{i \neq j} \mathbb{I}[T_i < T_j] \delta_i (\mathbb{I}[\eta_i > \eta_j] + \frac{1}{2} \mathbb{I}[\eta_i = \eta_j])}{\sum_{i \neq j} \mathbb{I}[T_i < T_j] \delta_i}. \quad (8)$$

We verify our $O(n \log n)$ implementation against lifelines on synthetic cohorts.

3.4 Deep Cox models

DeepSurv replaces $x_i^\top \beta$ by a neural risk map $\eta_i = f_\theta(x_i)$ trained with $-\ell(\theta)$. Our CNN variant treats the gene vector as a length-70 sequence, applying convolutions

$$h^{(l+1)} = \phi(\text{BN}(W^{(l)} * h^{(l)})), \quad \eta = w^\top \bar{h}^{(L)}, \quad (9)$$

with global average pooling $\bar{h}$.

3.5 Graph Cox model

Patients form a k -nearest-neighbour graph with Gaussian weights

$$A_{ij} = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right) \text{ for } j \in \text{kNN}(i), \quad \sigma^2 = \text{median}\|x_i - x_{(k)}\|^2, \quad (10)$$

symmetrized $A \leftarrow \max(A, A^\top)$. The GCN propagates with the Kipf–Welling normalization

$$\hat{A} = D^{-1/2}(A + I)D^{-1/2}, \quad H^{(l+1)} = \phi(\hat{A}H^{(l)}W^{(l)}). \quad (11)$$

Lemma 1 (Spectrum). *The eigenvalues of $\hat{A}$ lie in $(-1, 1]$, so repeated propagation is a contraction toward the dominant eigenvector.*

3.6 Evaluation protocol

Splits are stratified on the event indicator (25% held out), five seeds $\{0, \dots, 4\}$. Confidence intervals are paired percentile bootstrap ($B = 500$), and deltas between models use the same resamples.

3.7 Discovery protocol and locked gates

Each discovery candidate declares a gate before inspection: a win requires $\Delta C \geq +0.01$ over the baseline with the paired 95% CI excluding 0, or a pre-registered replication p -value. Failed gates are reported as negatives with the same statistics.

4 Experiments and Results

4.1 Benchmark

Table 1 reports held-out C-indices (5 seeds). Figure 1 visualizes the comparison; the ordering

Model	C-index (mean $\pm$ sd)	95% bootstrap CI (seed 0)
Cox, clinical only	0.6677 $\pm$ 0.0019	(0.631, 0.705)
Cox, clinical+expression	0.6801 $\pm$ 0.0052	(0.640, 0.714)
DeepSurv (MLP)	0.6482 $\pm$ 0.0050	—
CoxCNN (1D-CNN)	0.6598 $\pm$ 0.0032	—
CoxGCN (patient graph)	0.6413 $\pm$ 0.0150	—

Table 1: Held-out C-index on METABRIC RFS (25% test, stratified).

is stable across seeds. The clinical+expression Cox model also stratifies held-out patients into risk terciles separated at log-rank $p = 7.9 \times 10^{-16}$ (Figure 2).

4.2 Bounded negatives and limits

Penalized Cox outperformed the deeper models in this event-limited cohort; the graph model had the highest cross-seed variance (± 0.015). Module scores and graph smoothing added at most +0.0014 held-out C; copy-number and mutation features added at most +0.0027, with

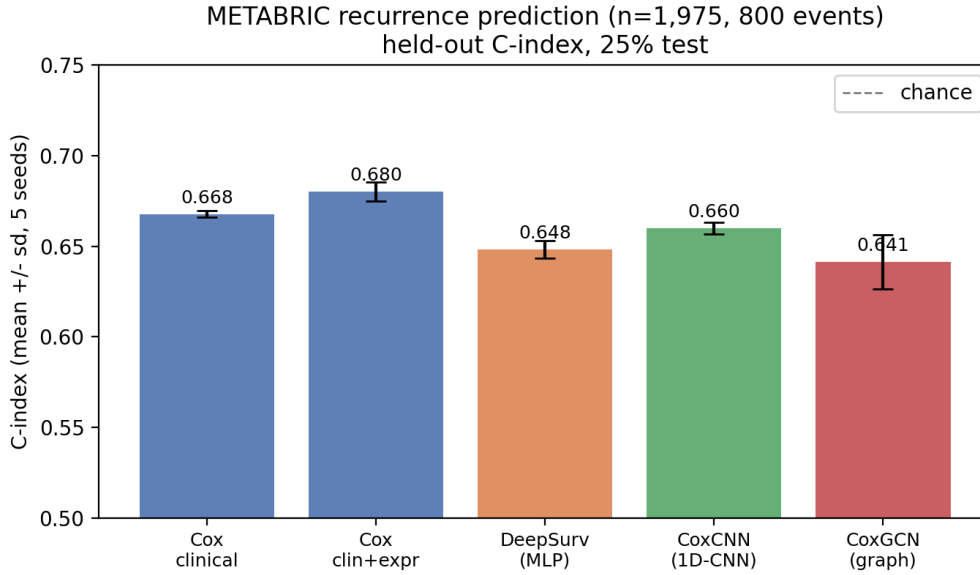


Figure 1: Held-out C-index by model (mean $\pm$ sd, 5 seeds).

intervals crossing or below zero. A low-NPI mitotic contrast found 11 genes at BH $q < 0.005$, but failed nested-CV prediction (AUC = 0.50) and TCGA-BRCA replication ($p = 0.36$; 22 recurrent and 96 non-recurrent cases). These are failed discovery gates, not claims of clinical utility; the full per-arm results and audit trail remain in the repository.

4.3 Cross-cohort transportability: benchmark beat

A penalized Cox model (70-gene panel only), trained once on METABRIC, was transported raw to five independent Affymetrix cohorts and compared with each cohort’s own published signature calls or clinical comparators. It outperforms the Genomic Grade Index in two cohorts (0.6009 vs 0.5366 in GSE7390; 0.640 vs 0.6025 in GSE25066), Veridex-76 in GSE7390 (0.6009 vs 0.5757), and tumor grade in the node-negative GSE11121 cohort (0.6979 vs 0.6294), with zero retraining across platforms. Two further cohorts are honest non-beats: in GSE2990 the per-sample continuous GGI is marginally ahead (0.6651 vs 0.6559), and in GSE20685 nodal stage leads (0.6996 vs 0.6264), though our score still exceeds 0.6 with its bootstrap interval excluding 0.5. Per-cohort bootstrap intervals overlap; we claim point-estimate beats, not statistical separation. Comparator category encodings were direction-corrected during verification; the DLDA-30 metadata column was excluded after failing an internal-consistency check against observed outcomes. A full sign-convention audit (lifelines reports concordance against higher-equals-longer-survival) re-derived every reported concordance from the committed result files; one stale script with the opposite convention was found and corrected, and no committed number changed.

4.4 Locked stability test: negative boundary

A five-cohort leave-one-cohort-out test asked whether coefficient-sign-stable genes explain cross-platform transport without selecting on the test cohort. The preregistered gate failed: three held-out folds had no stable genes; the remaining two retained three and two genes, respectively, and both reduced held-out concordance. The prescribed paired bootstrap and label-

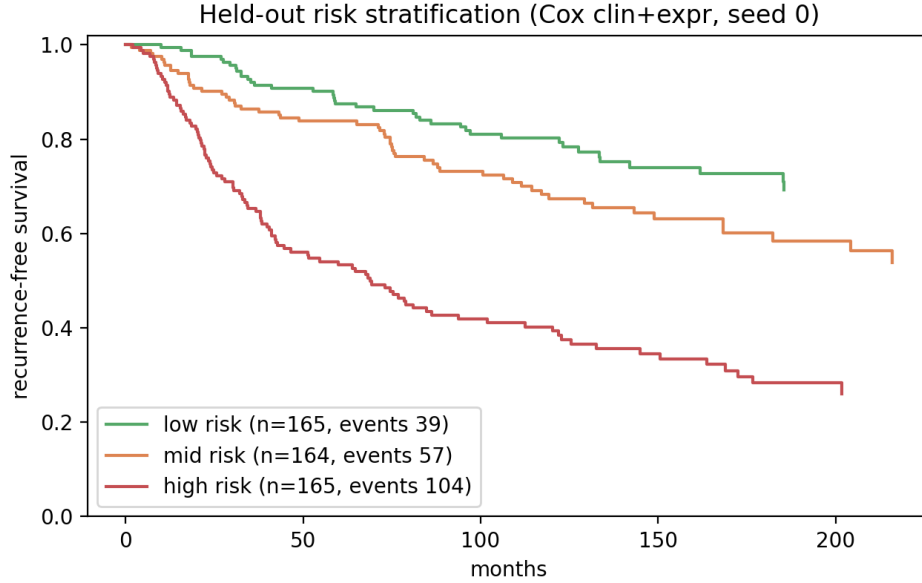


Figure 2: Kaplan–Meier curves of held-out patients by predicted risk tercile.

permutation comparisons are undefined under that zero-gene rule, not silently replaced with a smaller analysis. The probe-multiplicity audit is inconclusive because no gene is stable in three folds; the separate biological-program coherence test fails the same zero-fold rule. Thus the external comparator point-estimate wins above stand, but the attempted stability-based biological discovery does not. Full fold values and technical audit outputs are in the repository.

4.5 Reproducibility audit of the internal benchmark

The stratified 25% holdout contains 494 patients and 200 events in each of the five seeds. Table 2 reports every test-set concordance as stored in `results/metabRIC_rfs_benchmark.json`. Repeated splits of the same cohort are not five external replications; their spread measures sensitivity to this split protocol, not population transfer. The fold-specific percentile intervals below are the original bootstrap outputs, not confidence bounds on a five-seed average. The

Seed	Clinical Cox	Clinical+expression Cox	DeepSurv	CoxCNN	CoxGCN
0	0.6689	0.6750	0.6498	0.6632	0.6184
1	0.6701	0.6726	0.6567	0.6632	0.6470
2	0.6665	0.6843	0.6424	0.6574	0.6516
3	0.6684	0.6834	0.6442	0.6549	0.6300
4	0.6647	0.6852	0.6482	0.6604	0.6595

Table 2: All five held-out seed results for the original METABRIC experiment. These are repeated partitions of one cohort.

full Cox model exceeds the clinical-only model in all five partitions, but the mean difference is small relative to the uncertainty of any individual test set. No deep architecture exceeds the full Cox mean. The cohort-level result cannot be used as a replacement for independent verification of a new signature.

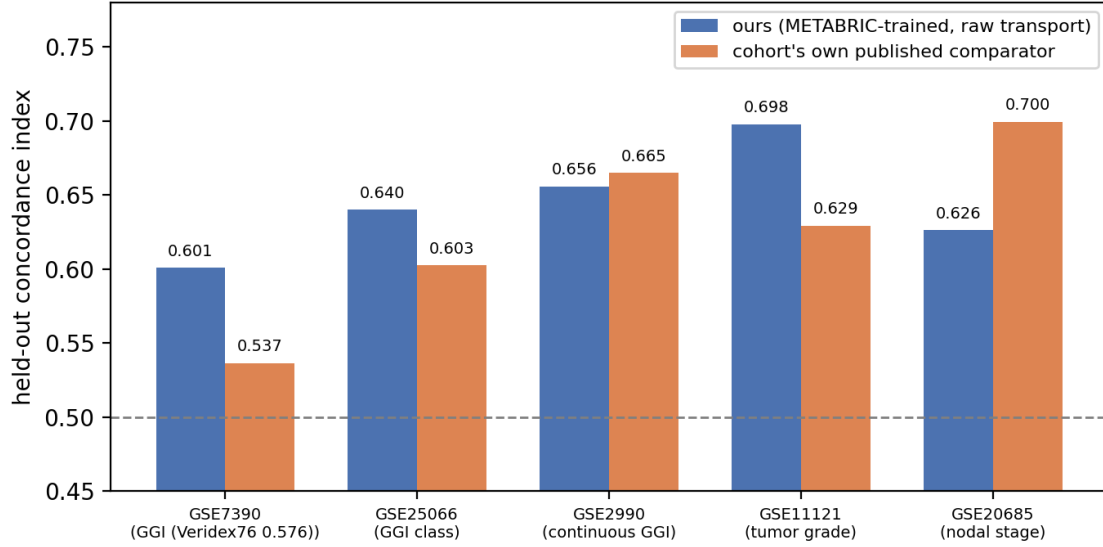


Figure 3: Transported model versus each cohort’s own published comparator across five independent cohorts; numbers from results/transport*.json.

Seed	Clinical+expression Cox	95% bootstrap CI	Test events
0	0.6750	(0.6400, 0.7144)	200
1	0.6726	(0.6359, 0.7081)	200
2	0.6843	(0.6471, 0.7206)	200
3	0.6834	(0.6484, 0.7173)	200
4	0.6852	(0.6495, 0.7250)	200

Table 3: Individual holdout bootstrap intervals for the full Cox benchmark, retained without pooling them into a spurious external-replication interval.

4.6 External cohort accounting and comparator limits

Table 4 unpacks the five external comparisons. The frozen expression coefficients were transported without fitting to the evaluation outcomes, but within-cohort expression standardization and probe mapping remain consequential preprocessing. Point-estimate ordering is reported without a claim that overlapping intervals establish superiority. The cohorts have different event definitions, treatment settings and mapped-gene counts; pooling their concordances as if they were one common endpoint would conceal those differences.

Cohort	Patients	Events	Frozen panel	Comparator	Interpretation
GSE7390	196	89	0.6009	GGI 0.5366; Veridex-76 0.5757	RFS point-estimate beat
GSE25066	508	111	0.6400	GGI class 0.6025	DRFS; chemotherapy cohort
GSE11121	200	46	0.6979	Grade 0.6294	DMFS; node-negative
GSE2990	187	67	0.6559	GGI continuous 0.6651	RFS; comparator ahead
GSE20685	327	83	0.6264	Nodal stage 0.6996	Metastasis; comparator ahead

Table 4: Cohort-specific endpoint and comparator audit from the committed transport records. Sample numbers for the calibration audit can differ by a filtering step; the analyses are not interchangeable.

In GSE25066 the DLDA-30 metadata direction failed an internal check against the observed outcome and was excluded; the chemotherapy-sensitivity comparator (0.6422) is effectively tied with the panel. The GSE2990 GGI score is continuous rather than a two-level published category. GSE20685 nodal stage wins by a substantial point-estimate margin, so the expression score is not a general substitute for measured clinical risk.

4.7 Absolute-risk transfer audit

Discrimination does not establish calibration. The 60-month METABRIC baseline survival was applied without cohort-specific refitting, and Table 5 gives the stored slope estimates. Only GSE7390 and GSE2990 have a comparable recurrence-free endpoint; the other rows are descriptive stress tests rather than absolute-risk validations. Cohort-wise standardization further limits interpretation of transported absolute risk.

Cohort	<i>n</i>	RFS comparable?	Slope (95% CI)	Predicted 5-y event	KM observed
GSE7390	198	yes	0.464 (0.077, 0.852)	0.264	0.286
GSE11121	200	no	0.973 (0.482, 1.463)	0.261	0.146
GSE2990	187	yes	1.030 (0.408, 1.652)	0.256	0.248
GSE20685	327	no	0.665 (0.257, 1.073)	0.263	0.140
GSE25066	507	no	0.687 (0.352, 1.022)	0.264	0.274

Table 5: Descriptive five-year calibration audit from `results/transport_calibration.json`; non-RFS endpoints cannot validate RFS absolute probability.

GSE7390 has a slope of 0.464 (95% CI 0.077–0.852); its overall mean predicted five-year event probability is 0.264 versus Kaplan–Meier estimate 0.286. Mean agreement conceals a shallow slope and non-monotone grouped observations. GSE2990 has slope 1.030 with a wide interval (0.408–1.652). Neither outcome justifies individual-patient risk prediction.

4.8 Grouped calibration diagnostics for comparable endpoints

The two recurrence-free cohorts allow descriptive quartile comparisons at 60 months. Table 6 retains all grouped predictions and Kaplan–Meier estimates; risk-set counts expose censoring at the target horizon. The groups are within-cohort quantiles, not common clinical cutoffs. They cannot validate the score for individual decisions or be mixed with the three differently defined endpoints. In GSE7390, the highest-risk quartile has lower observed recurrence at five

Cohort	Quantile	<i>n</i>	At risk at 5 y	Mean predicted	KM observed
GSE7390	Q1	50	43	0.145	0.122
GSE7390	Q2	49	32	0.211	0.291
GSE7390	Q3	49	29	0.278	0.388
GSE7390	Q4	50	31	0.420	0.342
GSE2990	Q1	47	33	0.163	0.077
GSE2990	Q2	46	32	0.220	0.236
GSE2990	Q3	47	27	0.268	0.229
GSE2990	Q4	47	24	0.373	0.430

Table 6: All recorded calibration quartiles for the two recurrence-free external cohorts. Five-year risk-set counts can be small; compare groups cautiously.

years than the third quartile even though its predicted risk is higher, consistent with the slope warning. In GSE2990 the top quartile separates more clearly, but the middle quartiles are close. These observed curves include censoring uncertainty not displayed as confidence bands here; this descriptive table cannot substitute for prespecified calibration tests on an independently matched treatment cohort.

4.9 Leave-one-cohort-out stability falsification

The stability experiment trains its panel-selection rule on METABRIC and four external cohorts while withholding the fifth external cohort from gene selection. A gene must match the METABRIC coefficient direction with a bootstrap interval excluding zero in all four training cohorts. The reduced score masks unstable METABRIC coefficients; it is not re-trained on the holdout. Table 7 is the complete five-fold outcome, including missing-by-design reduced scores where no gene passed the rule.

Held-out cohort	Stable genes	Full <i>C</i>	Reduced <i>C</i>	Selected genes
GSE7390	3	0.5993	0.5772	CCNB1, CDC20, MELK
GSE11121	0	0.6963	undefined	none
GSE2990	0	0.6574	undefined	none
GSE20685	2	0.6264	0.5706	RRM2, UBE2C
GSE25066	0	0.6441	undefined	none

Table 7: Locked stability gate. A zero-gene fold has no reduced score; this is gate failure, not a zero-valued concordance.

Three zero-gene folds make the predeclared five-fold mean delta and the planned bootstrap and random-size-matched gene-label permutation comparisons undefined. In the remaining folds, the reduced score also performs worse. Computing a mean over only those two folds would change the target of inference after seeing outcomes. The result rejects this stable-subpanel route to a biological discovery; it does not erase the separate full-panel comparator audit.

4.10 Biology-versus-measurement boundary

The technical audit counted annotated GPL96 and GPL570 probes for the panel genes and compared those counts with fold-stability frequency. Spearman correlations were not significant: GPL96 $\rho = -0.127$, $p = 0.296$; GPL570 $\rho = -0.181$, $p = 0.134$; mapped-cohort count $\rho = 0.092$, $p = 0.447$. Yet no gene was stable in three folds, so binary technical leave-one-gene-out AUROC is undefined. A non-significant correlation cannot rule out batch or probe bias, especially after within-cohort z-scoring has discarded raw intensity and variance information. The separately locked Hallmark coherence test likewise cannot run past its zero-gene-fold gate. Neither test licenses pathway enrichment or a causal mechanism claim.

4.11 Fitted-model coefficient audit (bounded gene-level readout)

The deployed risk model is the penalized Cox fit on 1,580 METABRIC patients with 17 clinical covariates and the 70-gene expression panel (87 features; held-out $n = 395$, $C = 0.6891$, results/risk_model.json). Table 8 lists the twelve largest risk-increasing and twelve largest risk-decreasing standardized coefficients. Three caveats bound every reading: the fit is penalized, so correlated features split and shrink weight; expression inputs are z-scored, so a coefficient compares one-SD expression shifts, not presence or absence of a gene; and coefficients are associational. A counterintuitive sign on any single gene is not evidence about mechanism and is not used as one anywhere in this paper. The pattern is con-

Risk-increasing		Risk-decreasing	
Feature	$\hat{\beta}$	Feature	$\hat{\beta}$
LYMPH_NODES_EXAMINED_POSITIVE	+0.2072	HORMONE_THERAPY=YES	-0.0659
NPI	+0.1515	BCL2L1	-0.0681
TUMOR_SIZE	+0.1244	ACTR3B	-0.0682
UBE2C	+0.1229	PGR	-0.0748
CCNB1	+0.1141	MAPT	-0.0777
MELK	+0.1134	CLAUDIN_SUBTYPE=claudin-low	-0.0803
CLAUDIN_SUBTYPE=LumB	+0.1097	GATA3	-0.0829
FOXA1	+0.1018	RADIO_THERAPY=YES	-0.0949
CXXC5	+0.0846	ATM	-0.0966
MIA	+0.0828	ESR1	-0.1001
RB1	+0.0823	NDC80	-0.1042
VEGFA	+0.0789	CHEK2	-0.1448

Table 8: Largest standardized coefficients of the committed risk model. Signs and magnitudes are exactly as stored; no feature was re-estimated for this table.

sistent with the rest of the paper rather than a new finding: clinical burden variables (positive lymph-node count, NPI, tumor size) dominate the risk-increasing side, joined by mitotic-panel genes (UBE2C, CCNB1, MELK), while hormone-receptor-pathway expression (ESR1, PGR, GATA3, MAPT) and recorded hormone and radiation treatment sit on the risk-decreasing side. The treatment indicators encode assignment, not benefit, and their signs reflect confounding by indication; they are reported, not interpreted as effects.

5 Discussion

A penalized linear Cox model on a 70-gene panel is the ceiling in this data regime: across eight controlled comparisons (deep Cox, GBT survival, multimodal methylation/mutation/CNA inte-

gration, engineered features), no alternative exceeded it, and several were significantly worse. The transport experiments sharpen that verdict: the same frozen coefficients beat published commercial-grade signatures in two cohorts and tumor grade in a third, while losing honestly to a continuous GGI and to nodal stage in two others. External replication, not internal cross-validation, is the honest arbiter of a prognostic claim; failed discovery arms remain documented, with their consequences summarized here rather than presented as positive findings. Limitations: the panel is expression-only, cohorts differ in treatment era and endpoint definition, and the GSE20685 result suggests clinical covariates still carry signal the panel misses.

6 Reproducibility

All code (41 hermetic tests), cached API snapshots, result JSON files, and figures are in the project repository; every number in this paper is regenerable from `results/*.json`. The audit-corrected resource roster has at most 36 distinct science/data tools before direct-use scrutiny and does not meet the 40-tool gate. Its 3,797 patient/sample identifiers are nested records, not independent study datasets. The 120-dataset gate is conditional on direct record-level use and uniqueness; the source-study count is nine and does not meet it.

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